

Prebuilt

Precision printing for labs that can't share their data

Printers learn their errors from each other. Patient scans, part geometry and measurements never leave the lab.

ScienceClaw · Manufacturing & Materials · MIT Media Lab

THE PROBLEM

Labs that print can't share what they learn

Guides, crowns and implants from patient scans

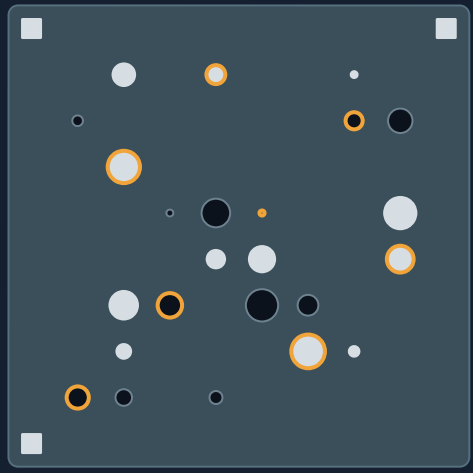
They must seat on the first print, and the scans are protected health data.

Microfluidic chips and bioprinted scaffolds

Sub-millimetre features, and every test print costs days and material.

Every printer is wrong in its own way

Holes print small and pins fat, by tens to hundreds of µm. Only printing reveals it.



The test plate: 24 pins and holes, 1–7 mm. Print it, measure it, know your printer.

Can printers learn from each other without sharing a single measurement?

BUILT ON PROVEN SCIENCE

Each piece is proven. Doing it privately is new

THE ERROR IS REAL

−250 µm

holes undersized and pins up to +195 µm on a desktop SLA printer, measured by CMM.

Tehnički vjesnik, 2026 (54 specimens, 5–20 mm)

CORRECTING IT WORKS

Model → fix

learn each printer's shape error, then print a compensated design.

Huang, Zhang, Sabbaghi, Dasgupta, IISE Transactions 47(5), 2015

SHARING HELPS

Data-poor

printers predict quality better by learning jointly with other printers.

Wei, Wang, Dickens, Chi, IISE Transactions 55(4), 2023; Sabbaghi & Huang, Ann. Appl. Stat. 12(4), 2018

The gap: cross-printer learning so far pools raw data or moves models centrally, and federated learning in printing covers defect images (Tang et al., RCIM 2024). Prebuilt shares three numbers, and nothing else leaves the lab.

HOW IT WORKS

Learn locally, share three numbers

1 Print a test plate

24 pins and holes. Measure them by micrometer or flatbed scanner.

2 Learn your printer

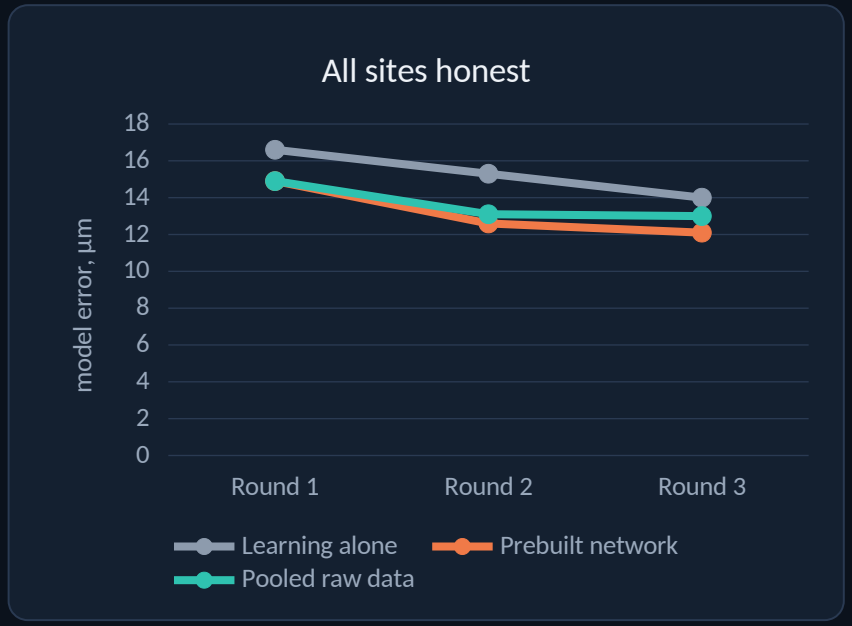
A physics model of your machine, fit locally on your own data.

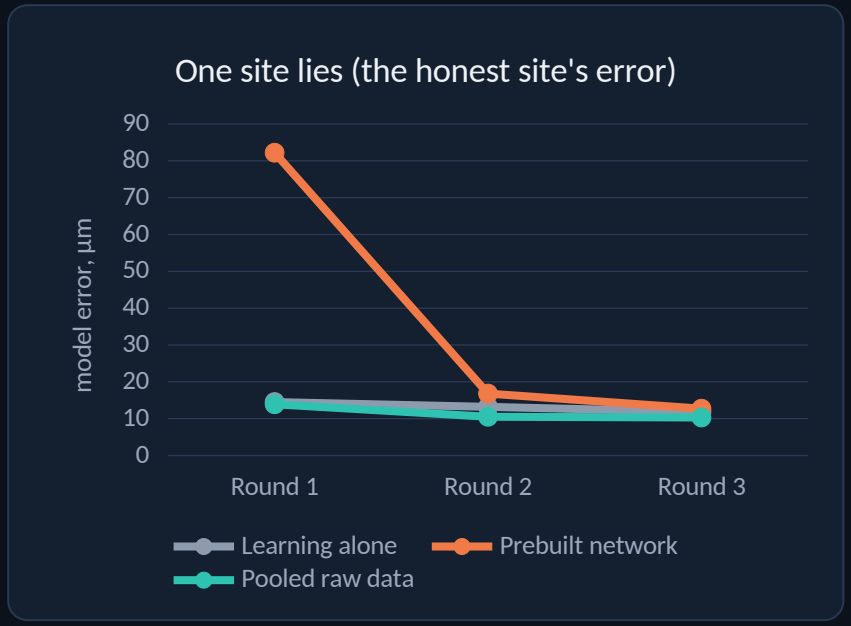
3 Share three numbers

Mean and spread of 3 parameters every printer shares. Schema-checked, logged on a receipt.

Never leaves the lab: patient scans, part geometry, measurements, machine settings. A replayable privacy audit checks every crossing: 0 measurements crossed.

The network learns faster, and shrugs off a liar





Average over 20 simulated fleets, registered config, error on held-out pieces. A liar spikes the honest site to 82.2 µm in round 1; trust collapses and it is back to 12.7 µm by round 3. With the consistency check on, round 1 stays at 13.3 µm.

RESULTS · SIMULATION

Parts that fit, and new labs get there faster

9 µm

size error from a corrected design, against 132 µm drawn as-is. For scale: crown margins must be under ~120 µm.

35%

lower error for labs with little data when eight share. The network won 20 of 20 simulated fleets.

12%

lower error with just two labs, matching what pooling all raw data gives (won 75 of 100 fleets).

0 rounds

lost to a lying lab with the consistency check: 180 of 180 lies caught, 0 honest notes dropped.

Error: how far the predicted size of a pin or hole is from its true printed size, on held-out pieces. Each simulated fleet is run both ways and compared head to head. The gain comes from labs sharing, confirmed by ablation (benchmarks slide).

RIGOR

A real printer for scale, simulation to score

Registered first

Every claim and prediction was committed, with its hash on record, before the first print.

Negative control

When printers share no physics, sharing gains nothing. The gain isn't built into the scoring.

Scored against known truth

Simulation checks every prediction against the true printed size, which a real measurement can't: two micrometer readings of one pin differ by 65 µm.

REAL PRINTER · FORM STUDIO 1

1.44 mm

a 1.50 mm pin, by micrometer (two readings: 1.41 and 1.48)

- Inside the vendor's ±0.08 mm spec
- Inside the simulator, at its small edge
- Next: calibrate on the full 24-piece plate

WHO IT'S FOR

Medical and bio labs that print and can't pool data

Hospital 3D-printing labs

Surgical guides and anatomical models from patient scans. The scans stay in-house.

Dental labs

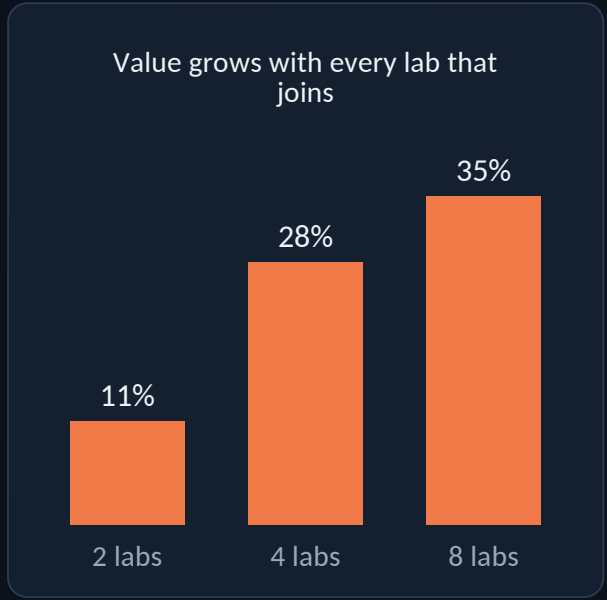
Crowns, models and aligners must seat first time. Patient data can't be pooled.

Bio and microfluidics labs

Lab-on-chip and organ-on-chip parts with sub-mm features. Recipes are IP.

Medical device makers

One part, many contract printers. Receipts double as an audit trail.



First markets we're targeting. Chart: lower error than learning alone at 2 / 4 / 8 data-poor labs, simulation. On sub-millimetre features the gain holds at 7–9% in simulation. Next: bioprinting.

The network beats every single-lab baseline

Round 3, simulated fleets, head to head	Lower error	Fleets where it wins	p
Network vs single site alone	12.3% (1.6 µm)	75 / 100	<0.001
Pooled raw data (upper bound) vs alone	12.2%	64 / 100	
Ablation: sharing off, same agents	2.4%	56 / 100	0.27
Ablation: agents vs random test pieces	+1.0%	31 / 60	0.90
4 / 8 data-poor sites vs alone	28.5% / 34.7%	35/40 · 20/20	<0.001
Negative control: nothing shared	−13.0% (adaptive spread −0.4%)	19 / 60	0.006

Hidden-truth simulator, registered config, error against true size on held-out pieces, sign test over worlds. 90% intervals cover 90–91%. GPU posterior up to 84× NumPy on an H100. Reproduce: docs/validation-and-benchmarks.md.



Print anywhere, trust the part

Labs learn their printers together. Their data stays home.

github.com/matte-realize/prebuilt · all sharing results in simulation